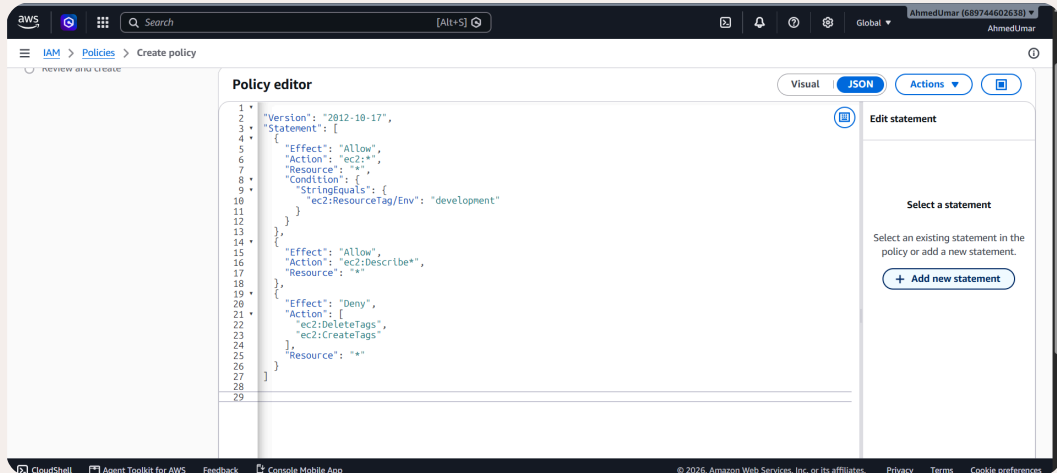


CLOUD SECURITY WITH AWS IAM

By Ahmed Umar Rehman | September 2026



Introducing Today's Project!

Tools and concepts

Services I used were

AWS Alias

AWS IAM Users

AWS IAM Group

Key concepts I learnt include...

- How to make policies in json for users

Project reflection

This project took me approximately 2 hours The most challenging part was Alias Signin

Tags

What I did in this step

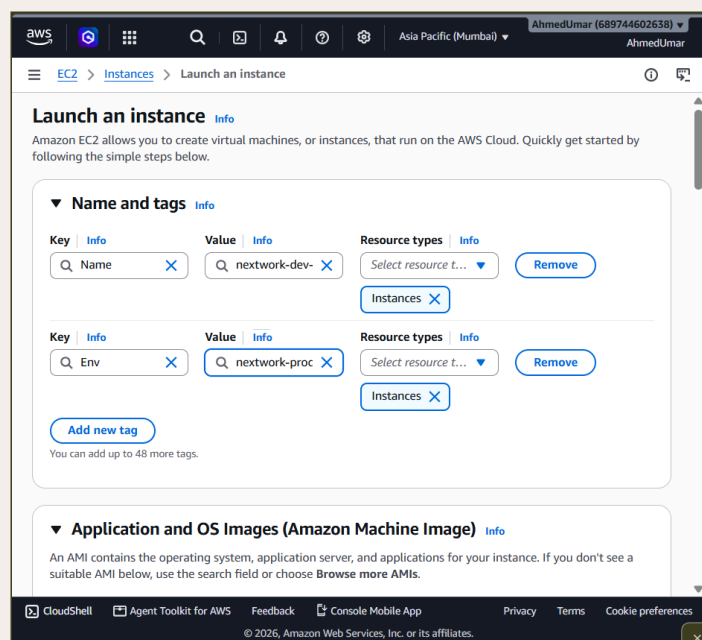
In this step, I will launch two EC2 instances to increase NextWork's computing power

Understanding tags

Tags are to remember which instance is used for

My tag configuration

The tag I've used on my EC2 instances is called nextwork-dev-ahmedumar3313 & nextwork-prod-ahmedumar3313 The value I've assigned for my instances are also same



IAM Policies

What I did in this step

In this step, I will create IAM Policy because i want to give access to development instance which I made

Understanding IAM policies

IAM Policies are rules about what to do about your resources who should you give and who should not

The policy I set up

For this project, I've set up a policy using JSON format

Policy effect

I've created a policy that user will not delete or create usertags of instances

Understanding Effect, Action, and Resource

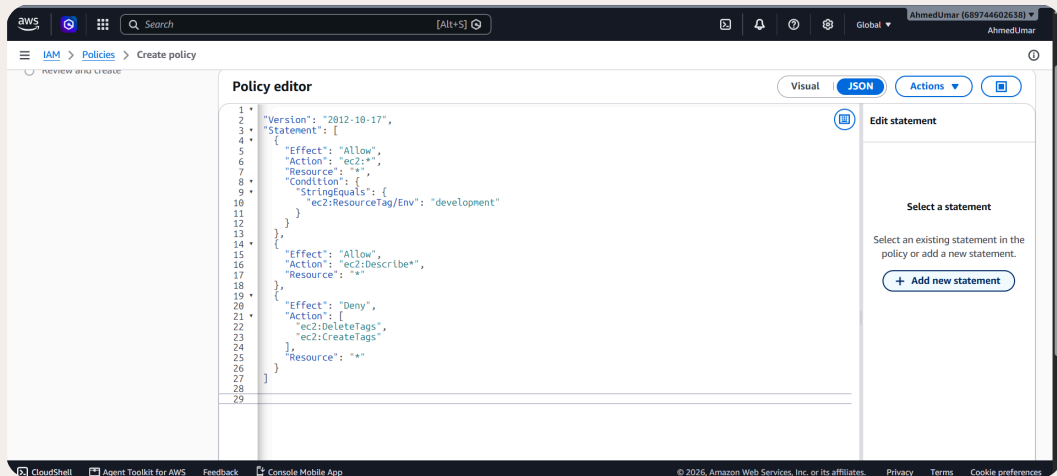
The Effect, Action, and Resource attributes of a JSON policy means

Effect: what to allow or what to deny

Action: what specific thing user can perform or not

Resource: Which AWS resource the action applies to

My JSON Policy



Account Alias

What I did in this step

In this step, I will simplify user login in AWS Account using Account Alias because it gives your account a custom, human-readable name used in the IAM sign-in URL,

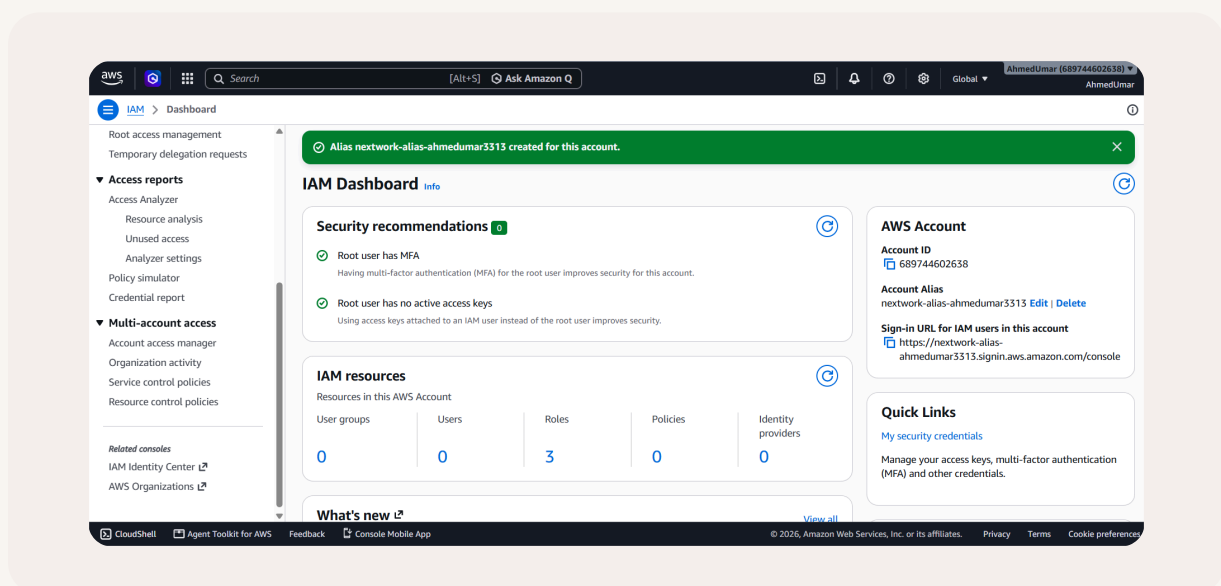
Understanding account aliases

An account alias is used to create easier user name login

Setting up my account alias

Creating an account alias took me exactly 10-12 minutes. Now, my new AWS console sign-in URL is

<https://nextwork-alias-ahmedumar3313.signin.aws.amazon.com/console>



IAM Users and User Groups

What I did in this step

In this step, I will Set up a dedicated IAM group & User because I want to see how can I manage all intern candidate permissions from one place

Understanding user groups

IAM user groups are those who get access to your resources and they also have limitations which resources to get and which to not

Attaching policies to user groups

I attached the policy I created to this user group, which means the user cannot delete or create user tags in EC2 instance which they will receive

Understanding IAM users

IAM users are those who represent one identity

Logging in as an IAM User

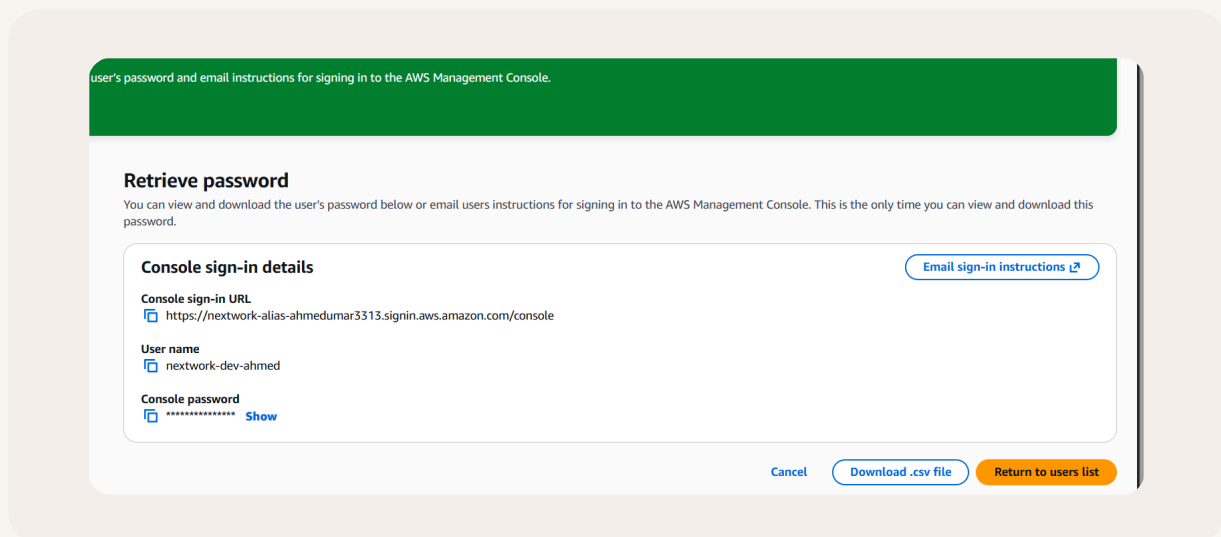
Sharing sign-in details

The first way is
download csv

The second way is
give user signin credentials

Observations from the IAM user dashboard

Once I logged in as my IAM user, I noticed I am seeing many access denied This was because i setup the policy that user will not create or delete instances



Testing IAM Policies

What I did in this step

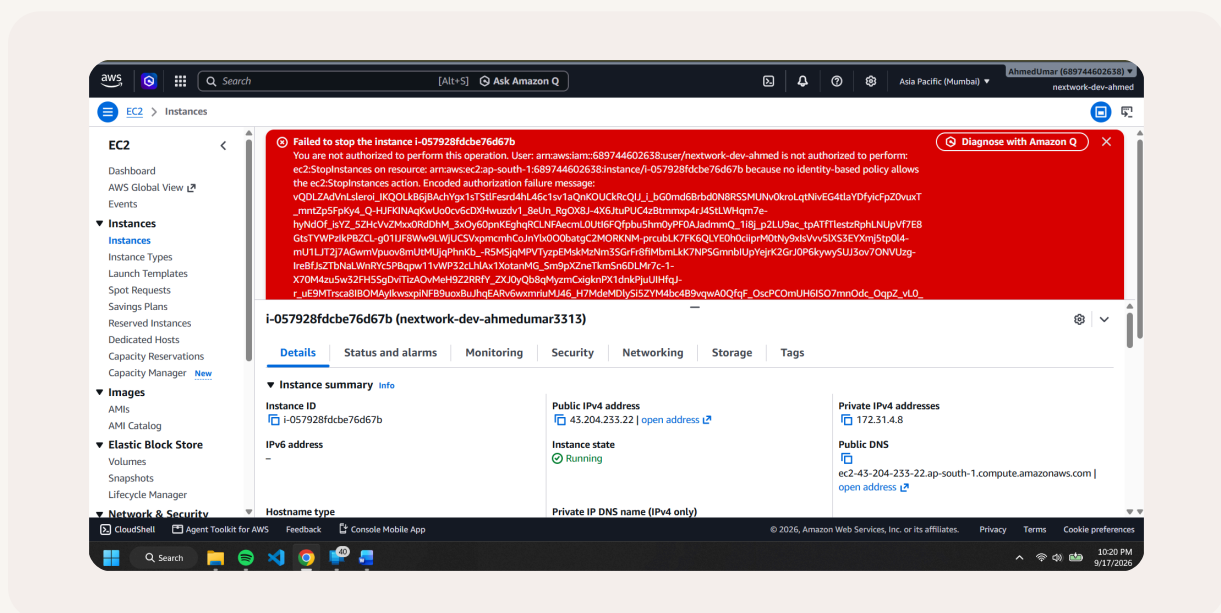
In this step, I will Log into AWS using the intern's IAM user. because it will test the intern's access to your production and development instance.

Testing policy actions

I tested my JSON IAM policy by not stop instances and it was successful

Stopping the production instance

When I tried to stop the production instance they told me that i was not authorized. This was because the IAM User has not permission to delete instances



Stopping the development instance

Next, when I tried to stop the development instance it showed me an error. This was because it was written in IAM Policies that i cannot stop instance

EC2

Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Instances (2)

Info

Last updated 2 minutes ago

Connect

Instance state

Actions

Launch Instances

Saved filter sets

Choose filter set

Filter Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☐	nextwork-dev-ahmedumar3313	i-057928fdce76d67b	Stopped	t3.micro	-	ap-south-1b	-
☐	nextwork-prod-ahmedumar3313	i-03852a7835b3bd51a	Running	t3.micro	3/5 checks passed	ap-south-1b	ec2-3-110-167-9

All instances

Instance alarm states

Access denied

View in CloudWatch

Service health

Access denied

View Service Health Dashboard

Health events affecting EC2 instances

Open events

None Show instances

Upcoming events

None Show instances

View AWS Health Dashboard

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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